Exploratory Data Analysis (EDA) Report – Geldium Delinquency Prediction Project

Introduction

This report presents an Exploratory Data Analysis (EDA) of Geldium's customer dataset. The objective is to evaluate data quality, identify missing values and anomalies, and determine key factors associated with customer delinquency.

Dataset Overview

The dataset contains **500 customer records** and **19 variables** consisting of demographic, financial, and payment history information.

Key Variables

- Age
- Income
- Credit Score
- Credit Utilization
- Missed Payments
- Loan Balance
- Debt-to-Income Ratio
- Employment Status
- Account Tenure
- Credit Card Type
- Location
- Monthly Payment History (Month_1 to Month_6)
- Delinquent Account (Target Variable)

Summary Statistics

- Average Age: **46.27 years**
- Average Income: **$108,379.89**
- Average Credit Score: **577.72**
- Average Credit Utilization: **49.14%**
- Average Missed Payments: **2.97**
- Average Loan Balance: **$48,654.43**
- Average Debt-to-Income Ratio: **29.89%**
- Average Account Tenure: **9.74 years**

Missing Data Analysis

The following missing values were identified:

Variable	Missing Values
Income	39
Credit Score	2
Loan Balance	29

Recommended Treatment

Median imputation is recommended for Income, Credit Score, and Loan Balance because financial variables may contain outliers that could distort the mean.

Data Quality Assessment

Duplicate Records

No duplicate customer records were identified.

Potential Anomalies

Several values require further validation:

- Credit Utilization exceeds 100% for some customers (maximum value = 102.58%).
- Very low and very high income values may require review.
- High loan balances and debt-to-income ratios should be examined for potential financial stress indicators.

Delinquency Distribution

The target variable distribution is as follows:

Category	Count
Non-Delinquent Customers	420
Delinquent Customers	80

Observation

Approximately **16%** of customers are delinquent, while **84%** are non-delinquent. This indicates a moderately imbalanced dataset.

Key Risk Indicators

Based on exploratory analysis and domain knowledge, the following variables are expected to be the strongest predictors of delinquency:

1. Missed Payments

Customers with a higher number of missed payments are more likely to become delinquent.

2. Credit Utilization

High utilization levels indicate potential financial stress and increased repayment risk.

3. Debt-to-Income Ratio

Customers with higher debt burdens relative to income may struggle to meet repayment obligations.

4. Credit Score

Lower credit scores generally indicate a higher probability of delinquency.

5. Employment Status

Employment stability influences repayment capability and financial resilience.

6. Recent Payment Behaviour

Payment history recorded in Month_1 through Month_6 can provide early warning signals for future delinquency.

AI and GenAI Usage

Example Prompts

- Identify missing values and potential data quality issues.
- Summarize key risk indicators associated with delinquency.
- Recommend suitable imputation techniques for missing financial variables.
- Highlight important variables for predictive modeling.

AI Contribution

AI tools assisted with:

- Dataset summarization
- Missing value identification
- Risk factor assessment
- Documentation support

All findings were reviewed and validated using analytical reasoning.

Conclusion and Next Steps

The dataset is generally well-structured and suitable for predictive modeling. Missing values were identified in Income, Credit Score, and Loan Balance and should be addressed through median imputation. The analysis suggests that Missed Payments, Credit Utilization, Debt-to-Income Ratio, Credit Score, Employment Status, and Recent Payment Behaviour are likely to be the most influential factors in predicting delinquency.

The next phase should focus on data preprocessing, feature engineering, and development of a machine learning model to identify customers at risk of missing future payments and enable proactive intervention strategies.